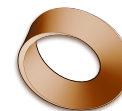


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.7 – Infrared

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can explain that all objects emit infrared radiation and that the amount depends on their temperature.
- b) I can describe how surface colour and texture affect the emission and absorption of infrared radiation.
- c) I can identify real-world applications of infrared technology, such as thermal imaging and remote controls.

Theory

Infrared radiation is a type of electromagnetic wave that sits just beyond the red end of the visible light spectrum. Although we cannot see it with our eyes, we can feel it as **heat**. Every object in the universe, from an ice cube to a burning star, emits infrared radiation. The hotter an object is, the more infrared radiation it emits. For example, a radiator in a school hallway emits much more infrared energy than the surrounding walls because it is at a higher temperature.

The way an object's surface looks also changes how it handles infrared radiation. **Dark, matt** (non-shiny) surfaces are excellent emitters and absorbers of infrared radiation. This means they lose heat quickly when hot and warm up quickly when exposed to heat sources. In contrast, **light, shiny** surfaces, such as polished metal or silvered thermos flask walls, are poor emitters and absorbers. They reflect infrared radiation rather than taking it in, which is why they are used to keep drinks hot or cold for longer periods.

Because infrared radiation is invisible, we need special tools to detect it. We can sense it on our skin as warmth, but to see it clearly, we use **thermal-imaging cameras**. These devices detect the infrared energy coming from objects and convert it into a visible image, often using colours to represent different temperatures (for example, white or red for hot areas and blue or black for cold areas). This technology is crucial for finding people trapped in rubble after an earthquake or locating swimmers in rough seas at night.

Infrared waves are also used in everyday household items. Television **remote controls** use pulses of infrared light to send signals to the TV without needing wires. Additionally, electric **heaters** often have shiny backs behind the heating element to reflect infrared radiation out into the room rather than letting it be absorbed by the wall, making them more efficient at warming the space.

Word bank: thermal-imaging cameras · dark · matt · light · shiny · remote controls · heaters

Activity

Investigating Surface Properties

Imagine you have four identical metal cans painted with different surfaces: 1. Black and matt 2. White and shiny 3. Silver and shiny 4. Black and shiny

You fill each can with hot water at the same starting temperature and leave them in a room for 20 minutes.

Complete the table below to predict which can will be the hottest and coldest at the end of the experiment. Remember: we want to keep the heat *in* the can (good absorber/emitter logic applies to losing heat to the environment as well).

Can Surface	Predicted Final Temperature (Hot/Cold)	Reason (Why?)
Black and matt		
White and shiny		
Silver and shiny		
Black and shiny		

Tip: Think about which surface would lose the most heat to the air around it.

Questions

1. Why can we feel infrared radiation as warmth even though we cannot see it?

2. Explain why a white car feels cooler than a black car on a sunny day, referring to absorption of radiation.

3. How does a thermal-imaging camera help firefighters find people in a burning building filled with smoke?

4. A scientist wants to design a satellite that stays cool in space. Should the outside of the satellite be covered in dark matt material or light shiny material? Justify your answer using the concepts of emission and absorption.

Extension

Find out how night-vision goggles differ from thermal-imaging cameras. Do they both detect infrared radiation, or do they work on different principles?
